

**GOVINDRAM SEKSARIA INSTITUTE OF
TECHNOLOGY & SCIENCE, INDORE
END SEMESTER EXAMINATION**

B.E. — First Year (Mechanical Engineering)
Maximum Marks: 56 Time: 3 Hours

SET E

UNIT I — Mechanical Properties (14 Marks)

- Q1.** (a) Define brittleness. (2 Marks) (b) Define fatigue limit. (2 Marks)
Q2. Explain ductile-to-brittle transition. (3 Marks)
Q3. Describe fatigue failure and S–N curve with neat diagram. (7 Marks)

UNIT II — Theory of Machines (14 Marks)

- Q4.** (a) Define simple mechanism. (2 Marks) (b) Define coupler link. (2 Marks)
Q5. Explain double slider crank mechanism with examples. (3 Marks)
Q6. Explain cam terminology and show diagrams for different follower motions. (7 Marks)

UNIT III — Fluid Kinematics (14 Marks)

- Q7.** (a) Define equipotential lines. (2 Marks) (b) Define discharge. (2 Marks)
Q8. Explain velocity potential with examples. (3 Marks)
Q9. Derive velocity components in terms of stream function. (7 Marks)

UNIT IV — IC Engines & Automobiles (14 Marks)

- Q10.** (a) Define brake specific fuel consumption. (2 Marks) (b) Define thermal efficiency. (2 Marks)
Q11. Explain knocking in SI engines. (3 Marks)
Q12. Explain valve timing diagram of a four-stroke SI engine with neat sketch. (7 Marks)